Exam #2 review

1) a) Use Midpoint, Trapezoid, and simpson's rule to approximate $\int_2^3 \frac{1}{\sqrt{x}}\,dx$, $n=4$. Find the error bounds for M_4.

$\Delta x = \frac{3-2}{4} = 1/4$

	17/8	19/8	21/8	23/8
2	9/4	10/4	11/4	3

$$M_4 = \frac{1}{4}\left(\frac{1}{\sqrt{17/18}} + \frac{1}{\sqrt{19/8}} + \frac{1}{\sqrt{21/8}} + \frac{1}{\sqrt{23/8}}\right)$$

$$T_4 = 1/4 \cdot \frac{1}{2}\left(\frac{1}{\sqrt{2}} + 2\frac{1}{\sqrt{9/4}} + 2\cdot\frac{1}{\sqrt{10/4}} + 2\cdot\frac{1}{\sqrt{11/4}} + \frac{1}{\sqrt{3}}\right)$$

$$S_4 = \frac{1}{12}\left[\frac{1}{\sqrt{2}} + 4\frac{1}{\sqrt{9/4}} + 2\frac{1}{\sqrt{10/4}} + 4\frac{1}{\sqrt{11/4}} + \frac{1}{\sqrt{3}}\right]$$

$f(x) = x^{-1/2}$, $f'(x) = -1/2\, x^{-3/2}$, $f''(x) = 3/4\, x^{-5/2}$

$|f''(x)| \le K$, $K = 3/4\,(2)^{-5/2}$

largest on $[2,3]$

$$|E_M| \le \frac{K(b-a)^3}{24n^2} = \frac{3/4\,(2)^{-5/2}(1)^3}{24(4)^2} = \frac{3}{8(4)^4\sqrt{2}}$$

2) State: The geometric series test: $\sum_{n=0}^{\infty} a_n(r)^n$ is divergent if $|r| \ge 1$

Comparison test: $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \le b_n$, $\sum b_n$ is convergent.

Limit comparison test: $\sum a_n$, $\sum b_n$ are series with positive terms.

if $\lim_{n\to\infty} \frac{a_n}{b_n} = C$, and $C > 0$ and finite, then both series either converge or diverge together.

Test for divergence: If $\lim_{n\to\infty} a_n \ne 0$ then $\sum_{n=1}^{\infty} a_n$ is divergent.

Integral test: If f is positive, decreasing, and continuous on $[1,\infty)$, let $a_n = f(n)$

Then the series $\sum_{n=1}^{\infty} a_n$ is convergent iff $\int_a^b f(n)\,dn$ converges.

Ratio test if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L$, if $L<1 \Rightarrow \sum a_n$ is absolutely convergent

$L>1 \Rightarrow \sum a_n$ is divergent

$L=1 \Rightarrow$ test is inconclusive

Root test if $\lim \sqrt[n]{|a_n|} = L$, if $L<1 \Rightarrow \sum a_n$ is absolutely convergent

$L>1 \Rightarrow \sum a_n$ diverges

$L=1 \Rightarrow$ test is inconclusive.

Alternating series Test: if $b_{n+1} \le b_n$ for all n and $\lim_{n\to\infty} b_n \neq 0$

then $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ Diverges.

if $\lim b_n = 0$, $|R_n| = |S - S_n| \le b_{n+1}$

P-series test: $\sum_{n=1}^{\infty} 1/n^p$ is divergent if $p \le 1$ else if $p>1$, it converges

if R_n is the remainder estimate for the integral test

$R_n = S - S_n$, $\int_{n+1}^{\infty} f(n) \le R_n \le \int_n^{\infty} f(n)$ where $f(n) = a_n$

3) a) find the length of the arc of the curve $x = \frac{y^4}{8} + \frac{1}{4y^2}$, $1 \le y \le 2$

$$L = \int_1^2 \sqrt{1 + \left(\frac{\partial x}{\partial y}\right)^2}$$

$$\frac{\partial x}{\partial y} = \frac{y^3}{2} - \frac{1}{2} y^{-3}$$

$$\left(\frac{\partial x}{\partial y}\right)^2 = \frac{1}{4} y^6 - \frac{1}{2} + \frac{1}{4y^6}$$

$$= \int_1^2 \sqrt{1 + \left(\frac{1}{4} y^6 - \frac{1}{2} + \frac{1}{4y^6}\right)}$$

$$= \int_1^2 \sqrt{\left(\frac{1}{2} y^3 + \frac{1}{2} y^{-3}\right)^2}$$

$$= \int_1^2 \left(\frac{y^3}{2} + \frac{y^{-3}}{2}\right) dy$$

$$= \frac{1}{8} y^4 - \frac{1}{4} y^2 \Big|_1^2 = 2 + \frac{1}{16}$$

3(b) Find the area of the surface created by rotating about the y-axis

$y^{2/3} + x^{2/3} = 1, \; 0 \le y \le 1$

$x = (1 - y^{2/3})^{3/2}$

$\frac{\partial x}{\partial y} = \frac{-\sqrt{1-y^{2/3}}}{y^{1/3}}$

$\left(\frac{\partial x}{\partial y}\right)^2 = \frac{1-y^{2/3}}{y^{2/3}}$

$S = 2\pi \int_0^1 x \sqrt{1 + \frac{(1-y^{2/3})}{y^{2/3}}}\, dy$

$= 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{1/y^{2/3}}\, dy$

$= 2\pi \int_0^1 (1-y^{2/3})^{3/2} \left(1/y^{1/3}\right)$

$u = 1 - y^{2/3}$

$du = -\frac{2}{3} y^{-1/3}\, dy$

$-\frac{3}{2} y^{1/3}\, du = dy$

$= 2\pi \int -\frac{3}{2} u^{3/2}\, du$

$= -2\pi \cdot \frac{3}{2} \cdot \frac{2}{5} u^{5/2}$

$= -2\pi \frac{3}{5} (1 - y^{2/3})^{5/2} \Big|_0^1$

$= \boxed{6\pi/5}$

4.1 Test the following series for convergence or divergence. Mention the test used.

a) $\sum_{k=1}^{\infty} \frac{k}{\sqrt{k^3+3k^2+1}}$

$\frac{k}{\sqrt{k^3+3k^2+1}} \le \frac{k}{\sqrt{k^3}} = \frac{1}{\sqrt{k}}$

Use the comparison test, take $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ ... diverges since $p \le 1$

b) $\sum_{n=1}^{\infty} \frac{n}{3^n + 5n}$

Use ratio test

$\lim_{n\to\infty} \left| \frac{n+1}{3^{n+1}+5(n+1)} \cdot \frac{3^n+5(n)}{n} \right| = \lim_{n\to\infty} \left|\frac{n+1}{n}\right| \cdot \lim_{n\to\infty} \left| \frac{3^n+5n}{3^{n+1}+5(n+1)} \right|$

$= 1 \cdot \frac{1}{3} < 0$ ... absolutely convergent

4 c) $\sum_{n=2}^{\infty} \frac{(-1)^n}{n\ln(n)}$

use the integral test on $\frac{1}{n\ln n}$

$$\int_2^{\infty} \frac{1}{n\ln(n)}\,dn = \lim_{t\to\infty}\int_2^{t} \frac{1}{n}\frac{1}{\ln(n)}\,dn$$

$u = \ln(n)$
$du = \frac{1}{n}dn$
$dn = n \cdot dn$

$$= \lim_{t\to\infty}\int 1/u\,du$$

$$= \lim_{t\to\infty} \ln(u)\Big|$$

$$= \lim_{t\to\infty} \ln(\ln(n))\Big|_2^t$$

$$= \lim_{t\to\infty} \ln(\ln t) - \ln[\ln(2)] = \infty$$ -- diverges

Alternating series test:

i $a_{n+1} = \frac{1}{(n+1)(\ln(n+1))} \leq \frac{1}{n\ln(n)} = a_n$

ii $\lim_{n\to\infty} \frac{1}{n\ln n} = 0$ $\therefore$ conditionally convergent.

d) $\sum_{n=1}^{\infty} \frac{(3\cdot5\cdot7\cdot9\cdots(2n+1))}{e^n n!}$

use ratio test

$$\lim_{n\to\infty}\left|\frac{3\cdot5\cdot7\cdots(2n+1)(2n+3)}{e^{n+1}(n+1)!}\cdot\frac{e^n n!}{3\cdot5\cdot7\cdots(2n+1)}\right| = \lim_{n\to\infty}\left|\frac{2n+3}{e(n+1)}\right| = 2/e < 1$$

Hence, it is absolutely convergent

5 Determine if the series is convergent. If it is, find the sum.

a) $\sum_{k=0}^{\infty} \frac{3^{k+1}}{5^{k+2}}$

$= \sum_{k=0}^{\infty} \frac{3^k 3}{5^k(25)} = \sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k \frac{3}{25}$, $|r| = 3/5 < 1$ converges --- Geometric series test

Geometric sum $\frac{a_0}{1-r} = \frac{3/25}{1-3/5} = \boxed{3/10}$

b) $\sum_{n=1}^{\infty} \frac{1}{n^2+3n}$

$\frac{1}{n^2+3n} = \frac{1}{n(n+3)} = \frac{A}{n} + \frac{B}{n+3}$

$A(n+3) + Bn = 1$

$(A+B)n + 3A = 1$, $A = 1/3$, $B = -1/3$

$\sum_{n=1}^{\infty} 1/n^2+3n = \sum_{n=1}^{\infty} \frac{1/3}{n} - \frac{1/3}{n+3}$

$= \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{9} + \cancel{\frac{1}{12}} + \cancel{\frac{1}{15}} \cdots\right) - \left(\cancel{\frac{1}{12}} + \cancel{\frac{1}{15}} + \cancel{\frac{1}{18}} \cdots\right)$ --telescoping sum

$= \boxed{\frac{1}{3} + \frac{1}{6} + \frac{1}{9}}$ convergent

6) Determine if the integral is convergent.

$$\int_1^3 \frac{x}{\sqrt{x^2-1}} = \lim_{t\to 1^+} \int_t^3 \frac{x}{\sqrt{x^2-1}}$$

$u = x^2-1$

$du = 2x\,dx$

$dx = \frac{du}{2x}$

$$\lim_{t\to 1^+} \int_t^3 \frac{x}{2x\sqrt{u}}\,du$$

$$\frac{1}{2}\lim_{t\to 1^+} \int_1^{\prime} \frac{1}{\sqrt{u}}\,du = \lim_{t\to 1^+} \sqrt{(x^2-1)}\Big]_t^3$$

$$= \lim_{t\to 1^+} \sqrt{8} - \sqrt{t^2-1}$$

$= \sqrt{8}$ ... the integral is convergent

7) Express $0.\overline{09}$ as a ratio of integers

$$\sum_{n=0}^{\infty} \frac{9}{100}\left(\frac{1}{100}\right)^n$$

$$\frac{a}{1-r} = \frac{9/100}{1-1/100} = 9/99 = 1/11$$

1(b) How large should n be, if we need M_4 to be accurate within 10^{-3}

$$10^{-3} \le \frac{3/4(2)^{-5/2}(1)}{24n^2}$$ from $|E_m| \le \frac{K(b-a)^3}{24(n)^2}$

$$n \ge \sqrt{\frac{(3/4(2)^{-5/2})1000}{24}}$$